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Syngene	Standard Operating Procedure	Department: Quality Assurance
Title: INVESTIGATION AND ROOT CAUSE ANALYSIS		
Document Number: SOP-GMP-QA-0066	Version Number: 7.0	Effective Date: 24-Feb-2026

Signature Manifestation

Responsibility	Department	Signed By	Date
Author	Quality Assurance - Gmp - Biologics	Sandesh Mr/QA/ SYNGENE	23-Jan-2026 15:56:12 (IST)
QA Review	Quality Assurance - Gmp - Biologics	Devendra Singh/GMP QA - Biologics/SYNGENE	23-Jan-2026 16:07:04 (IST)
QA Approval	Quality Assurance - Gmp - Biologics	Sathwik Punja/GMP QA-Biologics/SYNGENE	23-Jan-2026 18:47:40 (IST)

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Syngene	Standard Operating Procedure	DEPARTMENT: Quality Assurance

TITLE:INVESTIGATION AND ROOT CAUSE ANALYSIS

Document Number:	Version no.:	Effective Date:	Next Review Date:
SOP-GMP-QA-0066	7.0	24-Feb-2026	24-Feb-2028

1.0 OBJECTIVE:

To define the procedure to conduct investigation of non-conformities, identify the root cause, and report corrective and preventive actions.

2.0 SCOPE:

This procedure applies across Syngene GMP verticals (large and small molecules) and Central Support Functions: Corporate Quality Assurance (CQA), Internal quality audit & Continuous Improvement and Training Department.

3.0 RESPONSIBILITY:

Tasks	User (Investigator)	Investigation team (User, CFT Members, SMEs)	Quality Assurance (QA)	Corporate Quality Assurance (CQA)
Identify and define the investigation team	✓	X	X	X
Use appropriate tools to investigate and identify the Root Cause	✓	X	X	X
Prepare an investigation report within the specified timeframe.	✓	X	X	X
Propose appropriate corrective and preventive actions (CAPAs) to address the identified root cause.	✓	X	X	X
Coordinate with QA to implement remedial actions as part of the CQA surveillance check.	✓	X	X	X
Participate in the investigations.	✓	✓	✓	X
Participate in the investigations based on QA requirements.	X	X	X	✓

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Review of the investigation report.	X	✓	✓	✓
Approve the investigation report.	X	X	✓	X
Inform and/or seek approval from the client as necessary.	X	X	✓	X
Provide necessary documents to CQA (as necessary) to conduct surveillance checks.	X	X	✓	X
Propose remedial actions as part of the observations raised by the CQA team in the investigation report as per their surveillance check	X	X	✓	X
Ensure implementation of agreed remedial actions within committed timelines	X	X	✓	X
Review and assess closed investigations for adequacy as part of surveillance checks	X	X	X	✓
Review and approve the remedial actions proposed by QA as part of the surveillance check	X	X	X	✓

4.0 ABBREVIATIONS:

1.	CAPA	Corrective Action and Preventive Action
2.	COA	Certificate of Analysis
3.	EDMS	Electronic Document Management System
4.	EHSS	Environment, health, safety and sustainability
5.	EQMS	Electronic Quality Management System
6.	MSAT	Manufacturing Science and Technology
7.	OOS	Out of Specification
8.	OOT	Out of Trend
9.	PD	Process Development

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10.	PRD	Production
11.	QMS	Quality Management System
12.	RCA	Root Cause Analysis
13.	SME	Subject matter Expert

5.0 DEFINITION:

5.1	**Brainstorming**	The process in which the investigation team quickly generates as many ideas as possible on the possible cause of a specific non-conformity or unexpected outcome.
5.2	**Contributing Factor**	An element that is either necessary for a failure to occur (in addition to the root cause) or that amplifies the failure impact, but it would not have caused the failure by itself.
5.3	**Correction**	An action taken to eliminate detected nonconformity.
5.4	**Corrective Action**	Action taken to eliminate the cause of nonconformity or other undesirable situations, and to prevent its recurrence.
5.5	**Direct Cause**	The immediate cause that directly led to failure or unexpected conditions is often the last event in a chain of events that fails. The direct cause may also be classified as the immediate cause or proximate cause.
5.6	**Definitive Root Cause**	The scientifically conclusive cause, as determined by supporting data. Correcting the definitive root cause will prevent reoccurrence of the problem.
5.7	**Human Error**	The execution of a prohibited action, the failure to correctly perform the required action, or the misinterpretation of information essential to the correct execution of an action.
5.8	**Immediate Action**	An action taken to fix the impact of a problem quickly.
5.9	**Investigation**	A detailed examination and analysis of observed conditions and acceptance criteria are needed to determine the cause of failure, prescribe corrective actions required to resolve the effect of

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		failure, and identify preventive measures to prevent the occurrence of similar failures.
5.10	**Non-conformity**	The nonfulfillment of a specified requirement. (such as, OOS, OOT, Deviations, Customer complaints, product returns and recalls, Data Integrity incidents and any other non-conformance or systemic failures).
5.11	**Performance influencing factors**	Factors - the characteristics of the job, the individual, and/or the organization- influence human performance and result in human errors.
5.12	**Preventive Action**	Action taken to eliminate the cause of a potential nonconformity or other undesirable potential situation, and to prevent its occurrence.
5.13	**Probable Root cause**	The cause that has been scientifically rationalized as being the root cause, with a reasonable probability (but not scientifically conclusive), based on supporting data.
5.14	**Root Cause**	One or more underlying factors that caused an event, condition, or chain of events that led to a nonconformity or unexpected outcome. The root cause is a controllable factor, which, if corrected, shall prevent the problem from reoccurring.
5.15	**Root Cause Analysis (RCA)**	The process of discovering the root cause(s) of a problem to identify appropriate solutions. RCA is a systematic, structured approach to determine the root cause(s) using various strategies, tools, and techniques.
5.16	**Surveillance Check**	A systemic process to review and evaluate the closed investigations randomly and periodically ensuring the adequate capturing of key investigation elements.
5.17	**User**	The Owner of the process / procedure / operation / area.

6.0 EHSS: Not Applicable

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7.0 PROCEDURE:

7.1 Introduction:

7.1.1 Initiate investigation on the identification of a nonconformity or unexpected outcome and log it in the relevant QMS document (for example, deviation, OOS, OOT, or customer complaint).

7.1.2 The User's department, together with a team comprising members of the respective QA, CQA, and SMEs from relevant departments and other cross functional departments impacted by (or potentially connected with) the identified nonconformity or unexpected outcome, are responsible for investigating the situation.

7.1.3 Execute the investigation process in a structured, chronological order and simple manner.

7.1.4 Redundant information such as root cause, CAPA, impact assessment, etc. are unnecessary for parent and child investigations, instead, provide the PR reference wherever applicable.

7.1.5 Use clear, concise language with shorter words and sentences to enhance understanding. Avoid lengthy paragraphs and conversations. Use tables, graphs, and charts to illustrate data when necessary.

7.2 Description of the non-conformity or unexpected outcome with detailed background:

7.2.1 Describe and document the nonconformity/unexpected outcome clearly and concisely, by gathering the facts with a detailed background on What, Where, When, Who, Which and How.

7.2.2 Describe the details of product, process, equipment, documentation, validation, and/or object(s) affected by the nonconformity/unexpected outcome.

7.2.3 Document the specific details on date and time of the nonconformity/unexpected outcome, time of reporting along with the personnel (Employee IDs) who identified and involved in the issue,

7.2.4 Document the details of observed nonconformity and desired state (as per procedure/document/ process etc.)

7.2.5 Attach the photographs and scan copy of reference documents (relating to the issue) or document the reference number of records in the description as applicable.

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7.2.6 Use a flowchart to demonstrate the procedure/process details where the nonconformity/unexpected outcome is identified (as feasible).

7.3 Sequence of events:

7.3.1 Document the events chronologically and elaborately, including the date and time stamp (as applicable).

7.3.2 Document a detailed step-by-step key event of activity, specific to the nonconformity, which led to the discrepancy.

7.4 Pre-evaluation of the non-conformity/unexpected outcome:

7.4.1 **Initial/Preliminary Impact assessment:**

7.4.1.1 Undertake an initial/preliminary impact assessment of the nonconformity or unexpected outcome:

- Evaluate any potential impact on the product and verify the effectiveness of the immediate action (if any).
- Evaluate the currently available relevant information associated with the nonconformity or unexpected outcome. Include operations, documents, records, equipment, and materials that may provide information about the cause, in the investigation.
- Interview the personnel associated with the nonconformity or unexpected outcome. Record the interview details in FORM-GMP-QA-0507, as applicable.
- Also, consider the potential impact of the nonconformity or unexpected outcome on similar products, facilities, or materials as part of the assessment and document decisions and actions accordingly.

7.4.1.2 After assessing the affected process, product, or equipment, document the immediate decisions and actions. For example, the decision on holding the batches until the investigation and impact assessment concludes.

7.4.1.3 Ensure preliminary investigation conclusions are driven by the observed data rather than seeking data to fit a predetermined conclusion.

7.4.2 **Immediate action/correction:**

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7.4.2.1 Provide the defined immediate action/correction to minimize the impact on product quality or quality systems.

7.4.2.2 The proposed action should adequately restore compliance/stop further damage/impact against the non-conformance.

7.4.2.3 Justify in case immediate action/correction are not applicable.

7.4.3 **Historical check:**

7.4.3.1 Extend the investigation to identify similar occurrences in the previous 12-month period (at the minimum). Provide a rationale if the selected review period is less than 12 months.

7.4.3.2 Undertake a historical check by reviewing a wide range of relevant data sources concerning the nature of the nonconformity; for example, logbooks, Track Wise records, investigation reports, audit reports, control/reference sample verification, annual review records, and trends to the respective operational unit (OU).

7.4.3.3 Where the following processes are involved, consider similar historical occurrences in the current investigation, including for different products:

- Systemic or facility or equipment non-conformances
- The same analytical test method
- The same manufacturing process.

Otherwise, restrict the consideration of similar past occurrences to the same product.

7.4.3.4 Document the past event PR references if the historical checks identify similar nonconformities. It is not essential to include the detailed root cause/CAPA.

7.4.3.5 During the root cause analysis evaluation, if the root cause is like a prior one, document it along with its associated CAPAs. Evaluate the past root causes to determine the following:

- If the current investigation rules out prior root causes, it indicates that the CAPAs on the previous root causes effectively prevented recurrence.
- In case (at the end of the current investigation) the root cause is the same as an earlier root cause, reevaluate the CAPAs of the previous nonconformities and their implementation and effectiveness checks against the current RCA and proposed CAPA. This assists in understanding the aspects of the CAPA in the prior nonconformity that were inadequate and led to repeated failure.

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7.4.3.6 Use the historical information to provide recommendations for the current RCA, as applicable.

7.4.4 Investigation: Include the following in the investigation (but not limited to):

- Verify the effectiveness of any immediate actions taken.
- Discuss with relevant personnel (Interview).
- A review of training that was provided for the respective method/process.
- Review the raw materials used in the batch processing and the sequence of addition of manufacturing steps followed.
- Review the executed batch manufacturing record, analytical raw data, and supporting documents.
- Review the equipment cleaning checklists, logbooks, procedures, and practices.
- Review the qualification, preventive maintenance, or calibration of the associated equipment, instrument, or utility.
- Review the design and construction features of the relevant equipment or facility.
- Review the packaging, storage, and environmental monitoring conditions of associated materials.
- Review the environmental monitoring and temperature monitoring of process rooms.
- Summate the process sequences and discuss them with the development and production personnel involved in batch execution.
- In the case of product failures, verify the negative experiments conducted and their documentation as part of the product development report. In case of site transfers, verify the same/similar kinds of failures encountered in earlier site/s documented in the technology transfer dossier. Use the information to provide recommendations for the current RCA, as applicable.
- Undertake the RCA using appropriate tools and justification for the identified root cause (see 7.5.1).
- Evaluate the impact on the quality of other batches, projects, or activities.
- Decide on the disposition or action required for the affected batch(es), process(es), or activity(ies).

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- Create a CAPA plan (see 7.7).
- Form a conclusion.

7.4.5 Document the identified possible causes clearly and logically, with appropriate scientific justifications in the investigation report.

7.4.6 Where relevant, include the following evidence of investigations in the investigation report: photographs, instrument printouts, process flowcharts, checklists, photocopy of labels, analytical data of investigational analysis, copy of procedures or COA, etc.

7.5 Investigation:

7.5.1 **Root cause analysis (RCA):**

7.5.2 Conduct a detailed investigation to analyze the root cause(s) of nonconformity or unexpected outcomes. Use the following RCA tools, as appropriate, to carry out a thorough analysis:

- GEMBA and Data Gathering (see 7.5.3)
- Process Mapping and Gap Analysis (See 7.5.4)
- Brainstorming (see 7.5.5)
- Checklists (see 7.5.6)
- Why-Why Analysis (5 Why analysis) (see 7.5.7)
- Cause and Effect Analysis/Fishbone (Ishikawa) Diagram (see 7.5.8)
- Fault Tree Analysis (see 7.5.9)
- Pareto Charts (see 7.5.100)

7.5.3 **GEMBA & Data gathering:**

7.5.3.1 Take a GEMBA walk (where applicable) across the site deemed non-compliant to understand the exact area of non-compliance and analyze the situation. Document the outcome in the template FORM-GMP-QA-0504.

<u>Steps for effective Gemba walk:</u>

Step 1: Define Your Objectives

The Gemba Walk must possess an objective or a theme. Define the goals for your Gemba Walk.

Step 2: Prepare your team

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Before going on your Gemba Walk, prepare your team by informing them when the walk will happen, explaining the topic of discussion, and sharing the benefits of the walk.

Step 3: Focus on The Process

Focus on the process, not on people or identifying problematic workers. Remember: The Gemba Walk is not an employee evaluation - it is a productivity evaluation.

Step 4: Stick to The Value Stream

Analyse the processes and apply valuable practices. Similarly, identify areas ripe for optimization to increase value when determining that the procedure does not provide value or causes drawbacks.

Step 5: Write down Observations.

List down the observations from the Gemba Walk.

Step 6: Ask Questions

Ask precise and direct questions to gather valuable insights into the processes and potentially problematic areas, such as:

- Who?
- What?
- When?
- Where?
- Why?
- How?

Step 7: Keep Suggestions for a later time.

A Gemba walk acts as a tool for observation. The action comes after.

Step 8: Walk with Your Team

Gemba walks are more effective when the team is involved, especially when walking together with another department.

Step 9: Keep the schedule flexible.

Undertake the Gemba walks to address nonconformities promptly and as necessary.

Step 10: Return to the Gemba.

Conduct future Gemba walks as necessary and verify previous action items.

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7.5.4 **Process mapping and Gap analysis:**

7.5.4.1 Process mapping in the investigation of a nonconformity is a visual tool used to clearly understand how a process works, identify where the problem occurred, and spot potential root causes. Refer to the sample diagram below:

7.5.4.1.1 Define the Process Boundaries

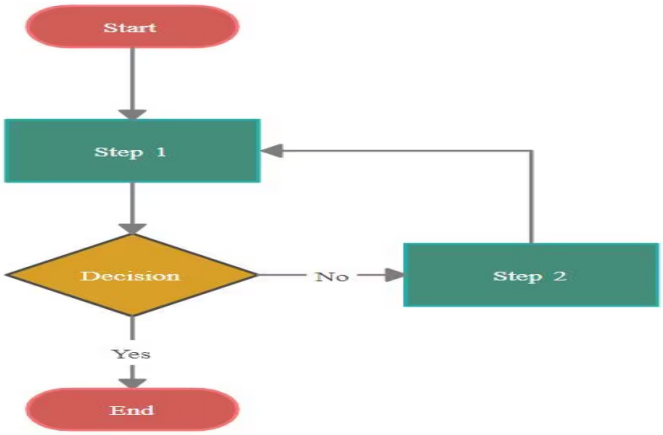
7.5.4.1.2 Map the Current ("As-Is") Process

7.5.4.1.3 Define the expected process (should be)

7.5.4.1.4 Analyse the process / Gap analysis.

7.5.4.1.5 Document the findings and root cause.

7.5.4.1.6 Propose CAPA.



7.5.4.2 Perform the Gap analysis to identify what happened and what did not happen as per procedure (what and where the non-compliance occurred, and the deviated step.)

7.5.4.3 List the functions tabularly (FORM-GMP-QA-0504), including what is required and what went wrong.

7.5.5 **Brainstorming:**

7.5.5.1 Use the brainstorming tool (FORM-GMP-QA-0504):

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- Users should invite SMEs and respective QA from different departments or functions (such as the investigation team) to participate in the brainstorming session on the nonconformity or unexpected outcomes.
- Based on their knowledge, each person should present probable reasons and scientific justification for the nonconformity or unexpected outcome.
- Capture the ideas of each participant.
- Narrow the discussions to questions on the nonconformity or unexpected outcomes.
- Limit the brainstorming session on time to ensure focused attention.
- Document probable causes from brainstorming and evaluation or analyzing possible root causes.

7.5.6 **Checklists:** Checklists are standard tools for investigations and RCA. They are designed to capture questions relevant to specific conditions.

7.5.6.1 Using the checklist tool:

- Include the common questions that need answering to identify the root cause of the nonconformity or unexpected outcome.
- Use questions that can be answered with “yes”, “no” or “not applicable”; either by means of tick boxes or by writing or indicating the answer.
- For each answer (“yes,” “no,” or “not applicable”), define the respective actions.
- Use the pre-defined checklists in place for investigations, such as OOS results (SOP-GMP-QA-0077) and OOT results (SOP-GMP-QA-0078). These checklists are explicitly designed to investigate product impact. Use additional tools as per this procedure to support investigational aspects and RCA if they are not fully covered in the pre-defined checklists.

7.5.7 **Why-Why analysis (5 Why analysis):** A method of asking repeated questions to explore the cause-and-effect relationships underlying a particular problem. The investigator starts with a precise and focused problem statement, such as “Why did this happen?”, then takes the answer and further asks, “Why did this happen?” and so on as many times as needed to identify the real (root) cause behind the issue.

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7.5.7.1 Using the Why-Why analysis tool (FORM-GMP-QA-0504):

- Include the personnel involved in and/or associated with the issue.
- Keep asking “why?” as many times as necessary until identifying an actionable root cause, which, if corrected, shall prevent a similar nonconformity or unexpected outcome in the future.
- The answer to each “why?” may require further investigation into potential reasons.
- Individual “why?” questions might have multiple answers, leading to the branching off further questions and ultimately identifying multiple root causes.

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For example:

Why did *S. aureus* contaminate Zone A?

↓ ← Because an Operator caused the contamination.

Why did the Operator cause the contamination?

↓ ← Because the Operator was absent during training on plant hygiene.

Why was the Operator does not present during training?

↓ ← Because the Operator was on vacation.

Why was there no training follow-up when the Operator returned from vacation?

↓

Root cause: Because there is no procedure defined for the follow-up of missed training.

- In this example, the sequential "Why?" questions can stop at this point as an actionable root cause is identified. Correct it by drafting the missing procedure to prevent a similar problem.

7.5.8 **Cause and Effect analysis/Fishbone (Ishikawa) Diagram:** The Fishbone is a cause-effect analysis technique that is particularly useful for failures in complex processes that may have diverse potential causes for each event in a chain of issues leading to an eventual nonconformity. It uses a multi-question approach, like the Why-Why analysis tool (see 7.5.7), to narrow down factors that may have caused the failure and rule out those that did not.

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7.5.8.1 The undesirable effect is placed as the main spine and secondary spine are potential causes and contributing factors that could have affected (or caused) the unexpected effects.

7.5.8.2 Drawing the Fishbone Diagram tool:

- Add the non-conformance or unexpected effect to the right-hand end of the main spine.
- Arrange factors as secondary spines (typically the six factors: personnel (man), materials, machine, measurement, methods, and milieu, also called "6M")
- For each factor, arrange process parameters, inputs, and variations in the hierarchy as per FORM-GMP-QA-0505.
- Document the outcome from the evaluation of the 6M tool in FORM-GMP-QA-0504.
- Brainstorm all the possible causes of the problem (see 7.5.3). Ask, "What could have caused this problem?" Write each possible cause as a branch from the appropriate factor (secondary spine) on the Fishbone Diagram. Write causes in several places if they relate to several categories.
- Ask "What could have caused this problem?" about each written cause. Write sub-causes branching off the cause branches.
- Continue to ask these questions, generate deeper causes, and organize them under related causes or categories. It helps identify appropriate CAPAs.

7.5.9 **Fault tree analysis (FTA)** is a graphical tool used to explore the causes of system-level failures and to identify appropriate corrective actions. Use it to identify the potential cause behind the failure (besides mitigating risks) before they occur and investigate to identify root causes of problems in the existing systems.

7.5.9.1 The FTA tool evaluates system (or sub-system) failures one at a time but can combine multiple causes of failure by identifying causal chains. The results are pictorially represented as a tree of fault modes. At each level in the tree, combinations of fault modes are described using logical operators ("AND, OR" etc.).

7.5.9.2 Using the FTA tool:

- Draw a fault tree analysis diagram per FORM-GMP-QA-0508.
- Add the non-conformance or unexpected effect at the top of the tree diagram.

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- Add direct causes one level below the non-conformance.
- For each cause, assess whether it represents a basic cause (root cause) or is the result of one or more lower-level causes.
  - Draw circles around basic (root) causes that are not developed further.
  - Draw rectangles around potential causes.
  - Continue until the diagram contains only root causes at the lowest level of each branch.
  - Where more than one cause leads to the level above, use gate symbols to connect the branches to indicate whether these operate together [AND symbolized by OR symbolized by]or on their own.

7.5.10 **Pareto analysis:** The Pareto principle states that most effects (typically expressed as 80%) result from a small number of causes (typically expressed as 20%); for example, typically expressed as "80% of problems is traceable to 20% of the causes". The Pareto analysis tries to prioritize cause-effect analysis to identify the few root causes (or a particular root cause) of a nonconformity or unexpected result. The analysis involves creating a list of causes, later sorted into a table or chart based on frequency.

7.5.10.1 Using the Pareto analysis tool:

- Refer to FORM-GMP-QA-0506 for the tool application.
- Start by listing the potential root causes identified for the nonconformity or unexpected outcome (for example, by using the Why-Why analysis or other tools listed above).
- Decide the criteria for comparing the possible causes (for example, frequency, cost, or other consequences of occurrence).
- Collect additional data required for each chosen criterion.
- Sort the causes by descending scores for each criterion.
- Present the causes in a table (and optionally, a chart), in descending order of the chosen criterion, showing absolute and cumulative data for each reason. The cause with the highest score (for example, 40% of the consequences) is listed first, followed by the next

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(for example, 25% of the consequences), and the cumulative total (for example, 65% in this example).

7.5.11 **Using the RCA Tools Effectively:**

7.5.11.1 Initially, attempt to identify the direct or proximate cause using individual or combination investigation tools. The most appropriate tool in this initial RCA step is Cause and Effect Analysis (Fishbone Diagram; see 7.5.8). Use additional tools to further drill down from the direct or proximate cause to find the definitive root cause or probable root cause. Why-Why analysis is the most appropriate tool for this step (see 7.5.7).

7.5.11.2 RCA is highly effective when the tools are used collectively. The tools mentioned above are examples; other RCA or risk assessment tools (for example, Hazard and Operability Study [HAZOP] analysis or Failure Mode and Effects Analysis [FMEA]analysis), depending on the specific situation, are applicable.

7.5.11.3 Justify the identified probable or definite root cause(s) and contributing factors scientifically. If a definitive root cause is evident, it can be justified with or without tools but documented scientifically in the investigation report.

7.5.11.4 If the investigation identifies a system failure as a root cause or contributing factor, extend the same or start a new investigation for all other batches, processes, or facilities that might be associated with (or affected by) the nonconformity or unexpected outcome.

7.5.11.5 Primarily, the investigation should focus on the systems and processes relevant to the nonconformity or unexpected outcome rather than on the individuals involved. However, wherever human error is suspected or identified as a root cause or contributing factor, investigate further to identify:

- Whether the action was inadvertent (error) or deliberate (non-compliance)
- Any performance-influencing factors, if applicable
- The root cause of the related process, procedural or system-based errors or problems, is to allow for a systemic solution to prevent recurrence.
- The root cause of the related process, procedural or system-based errors or problems, is to allow for a systemic solution to prevent recurrence.

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7.5.11.6 If a definite root cause for the non-conformance or unexpected outcome is not identifiable, assume or hypothesis for the probable causes (i.e., the most likely root causes) with justification.

7.5.11.7 Experiments and/or examinations necessary to identify the root cause are executed as part of the investigation.

7.5.11.8 Document the identified definite or probable root cause(s) in the investigation report.

7.5.12 In cases where the root cause of nonconformity is attributable to individual error, and the individual responsible is no longer available (for example, leaving the organization or extended leave), take the following steps:

- Document the individual's unavailability in the investigation records.
- Utilize all available documentation and records to reconstruct the event and support the investigation.
- Assess for any systemic or procedure gap that may have contributed to or failed to prevent the error.
- Proceed to close the investigation based on the evidence gathered and on identifying the corrective actions.

7.5.13 **Examples of common root causes (not comprehensive):**

Error Type	Probable root causes behind the error
System-based errors	• Gaps in the training program or contents. • Lack of supervision and review process. • A gap in the preventive maintenance or calibration program. • A gap in the facility maintenance program. • Lack of resources to complete the operation as per written instructions. • Lack of traceability, resources, references, or required information.

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TITLE:INVESTIGATION AND ROOT CAUSE ANALYSIS		

Document Number: SOP-GMP-QA-0066	Version no.: 7.0	Effective Date: 24-Feb-2026	Next Review Date: 24-Feb-2028

Error Type	Probable root causes behind the error
Equipment/utility/instrument related errors	• Inappropriate assessment while selecting equipment for the process. • Operation beyond qualified parameters. • Improper design and layout of equipment for the process.
External factors	• Non-conductive environmental conditions. • Errors by service providers, contract manufacturers or testing labs. • Changes recommended by client. • Electronic system failure, network error, application error.
Procedural errors	• Insufficient information in the written procedure, missing required information. • The procedure is unsuitable for the activity and cannot be followed. • Typographical errors in a document issued for the execution. • Lack of traceability and references.
Process-based errors	• Lack of scale-up evaluation with process variables. • Lack of clarity in the technology transfer documents. • Unpredictable quality attributes changes are due to insufficient process development studies.

Human errors:	
Error Type	**Probable root causes of error**
Learning gap (inadequate understanding of testing or activity procedure as part of training)	• Insufficient training instructions.
Memory gap (unable to remember the complete information of the activity)	• Insufficient training frequency.

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Performance inconsistency (having knowledge but inconsistent performance or results; lack of supervision)	• Lack of clarity in the established procedures.
Application problem (knowing but making slips or transcription errors)	• Practical problems while implementing the approved procedure.
Decision (inappropriate decision when identifying a discrepancy)	• Lack of knowledge on activities that tie into making quality decisions.
Omission (sequentially performing the activity, but with one or more key steps missing)	• Operator fatigue due to extended shift.

7.5.14 **Interim Investigation Report:**

7.5.14.1 Prepare an interim report with available information, including interim controls (if applicable) and plans for further actions, in the following situations:

- If the investigation remains incomplete within the target timeline of closure of the document triggering the investigation (for example, deviation, OOS, OOT, customer complaint, etc.)
- Additional experiments are required to confirm the identified cause(s).
- The investigation is complete, and the root causes are identified, but further time is necessary for the QC to complete sample analysis and initiate batch release.

7.5.14.2 Number interim reports of an investigation sequentially (Interim Investigation Report 1, 2, 3, etc.).

7.5.14.3 Structure the final investigation report using compiled information from all interim investigations and any additional information.

7.6 Final Impact Assessment:

7.6.1 Perform an impact assessment on the batch(es), product(s), analyses, and/or activities associated with the event or if the outcome impacts the compliance status of established systems.

7.6.2 Extend the impact assessment to other batches of the same product and other products that may have been associated with the specific failure, nonconformity, or unexpected outcome.

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7.6.3 If necessary, undertake a detailed risk assessment per SOP (SOP-GMP-QA-0050), in addition to the impact assessment, to assess the possible consequences of failure of the manufactured batch (or product).

7.6.4 A typical impact assessment will encompass, but is not restricted to, the following content. The impact assessment will cover the relevant elements and mark the other elements as "not applicable".

- Impact on the quality of the product
- Impact on other products manufactured in the same facility.
- Impact on other batches, materials under storage, equipment, facility, systems, documents, or analytical data.
- Impact on Quality Management System and regulatory submission documents.
- Impact on the validated state of the product or process.
- Impact on preventive maintenance, calibration or qualification status of the equipment, utility, or facility.

7.6.5 Suppose the investigation indicates that other materials, products, or batches may be impacted; in that case, QA must put on hold the affected systems and batch(s) (including in-process materials, unreleased materials and previously released materials), with the approval of Head-QA or designee until the impact assessment completion.

7.7 Proposal for Corrective and preventive actions:

7.7.1 Propose appropriate CAPAs for each identified root cause (definite or probable) and/or contributing factor identified in the RCA. Assign a responsible department and target completion date to each action.

7.7.2 The QMS investigation team should take the input of the CAPA implementer, in case they are not a part of the investigation team, in proposing the CAPA. The implementer must evaluate the feasibility of the proposed CAPA before finalization.

7.7.3 Document the rationale and justification for each proposed action. Align each action to an identified root cause (definite or probable) or contributing factor to ensure no additional risk is introduced into current systems.

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7.8 Closure of investigation report:

7.8.1 Conclude the investigation by summarizing the nonconformity or unexpected outcome, investigation, RCA, impact assessment, and appropriate CAPAs clearly and concisely in the investigation report (see 7.11).

7.8.2 Use the outcome of the investigation to make decisions on batch disposition and provide appropriate rationale for each decision in the investigation report.

7.8.3 The investigation team members shall sign the agreed investigation report and QA shall approve the same.

7.8.4 QA reviewer/approver shall ensure that the client's signature approval is obtained in the investigation report before closing the record in the Trackwise system, if required by the quality agreement.

7.8.5 After the investigation report approval, initiate CAPA as per SOP, SOP-GMP-QA-0075 to track the action items.

7.8.6 After approval, enclose the investigation report with the attachments (attach the documents as referred in the investigation report) to the parent QMS document and proceed further for the final closure of the document.

NOTE: Do not close the investigation without identifying a definite or probable root cause.

7.9 CAPA Effectiveness Check:

7.9.1 Perform the CAPA effectiveness verification (post-implementation evaluation) for implemented CAPAs, as per Corrective and Preventive Action (SOP-GMP-QA-0075).

7.10 Investigation Surveillance:

7.10.1 The CQA must trace the number of investigations that closed in the previous month, based on Track Wise records, belonging to each Operating Unit.

7.10.2 CQA shall select closed investigations from each operating unit monthly as per the below-mentioned selection criteria (but not limited to):

- Critical deviations.
- Re-classified deviations as per the escalation matrix criteria.
- Human error-related root cause

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- Major customer complaints.
- Deviation related- Late phase, commercial phase, and major deviations.
- Late phase & commercial phase product-related investigations of OOS & OOT
- Recurrent OOS investigations of environmental monitoring and water analysis, etc.
- Deviations flagged during audits or regulatory inspections

  Note: Final classification shall be verified for correctness by CQA for all the closed investigations.

7.10.3 CQA shall review and assess the selected investigations for the adequacy of key requirements mentioned in this SOP and document the identified observations (if any) in FORM-GMP-QA-0733 (Template for observations of investigation surveillance) and share with respective QA for their acknowledgment and response.

Note: Use individual annexures for each investigation.

7.10.4 Upon agreement with CQA, the respective QA shall document the remedial actions against the observations in FORM-GMP-QA-0739 (Template for response for observations of investigation surveillance).

7.10.5 The identified observations by CQA and the corresponding remedial actions / responses by respective QA shall be routed through the EDMS procedure as per SOP defined template - i.e., FORM-GMP-QA-0733 and FORM-GMP-QA-0739.

7.10.6 In the event of a disaster, facility disruption, or system downtime that renders the EDMS non-operational or unusable, a manual form shall be used as per the defined templates - i.e., FORM-GMP-QA-0733 and FORM-GMP-QA-0739.

Note: QA shall re-open the investigation, if necessary, and document the details in the FORM-GMP-QA-0739. CQA shall review the re-opened investigation report.

7.10.7 QA shall ensure that the re-opened investigation is closed within 30 calendar days. During the re-opening of investigations, initiate additional CAPAs to address the observations, if applicable.

7.10.8 During the monthly CQA quality management review meetings, the surveillance check outcome is discussed, and appropriate actions are proposed to improve investigation adequacy continuously.

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7.10.9 The CQA shall conduct specific training for the User and QA on the key elements identified from the surveillance observations as learning for further improvement.

7.10.10 CQA shall conduct periodic training for personnel (involved in routine investigations) on the key adequacy elements of investigation as part of continuous capability building.

7.11 Contents of the investigation report:

7.11.1 Use the template FORM-GMP-QA-0504 as a basis for the structure of the Investigation Report. The template may be varied as needed, depending on the extent and outcome of the investigation.

Note: Refer to FORM-GMP-QA-0732 for guidance for preparing the investigation report with key adequacy elements.

7.12 Procedure for Documents Preparation in EDMS.

7.12.1 The respective author shall route the observations identified by CQA and the corresponding remedial actions/responses from the respective QA through the EDMS under the Category 3 document type.

7.12.2 New Content shall prepare by Clicking on the Import tab in the Welcome screen or click → Import → File from the menu or drag and drop the document from your computer into the Doc List widget.

7.12.3 The 'Import File' popup window appears.

7.12.4 Click on the "…" icon to the right corner of the 'Files to Import' box to browse a document. Locate the PDF document created, in the Import File wizard verify the document appearing in the 'Files to Import' box. Click Next > and fill in the details as below.

7.12.4.1 Select "Creation Profile" as "Executed Documents"

7.12.4.2 Select "Group" as "Checklists"

7.12.4.3 Select "Subgroup" as Executed Checklist" and Click Next>.

7.12.5 'Edit properties' Page is displayed, enter all the mandatory fields (Indicated by *) and any other fields if required.

Note: 1. Training is not required.

2. Under the traceability Section mention the details for "New/Revise reason" section as "New" for all the documents.

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7.12.6 Click Next>, system will import the document, edit, and save the changes. Click "Check in and Finish" Document is created with unique name, Auto filing, rendition (PDF format of the document) is auto created, status is set as 'Draft' and version is 0.1.

7.12.7 Author can self-approve the document or forward the document for Approval.

7.12.8 Approver approves the document, if approver rejects the task document moves to draft status, Author can import the new document by highlighting the existing document click on import and select new version. The document version number is incremented by 0.1

7.12.9 Once the document is approved the version is changed from minor to major as 1.0.

8.0 ANNEXURES:

Sr. No.	Annexure / Form / Template No.	Annexure / Form / Template title
1.	FORM-GMP-QA-0503	SOP Training Questionnaire
2.	FORM-GMP-QA-0504	Template for Investigation Report
3.	FORM-GMP-QA-0505	Template for Cause-and-Effect Analysis
4.	FORM-GMP-QA-0506	Template for Pareto Analysis
5.	FORM-GMP-QA-0507	Template for Interview Form
6.	FORM-GMP-QA-0508	Template for Fault Tree Analysis
7.	FORM-GMP-QA-0732	Guidance for preparing the investigation report with key adequacy elements
8.	FORM-GMP-QA-0733	Template for observations of investigation Surveillance
9.	FORM-GMP-QA-0739	Template for response for observations of Investigation

9.0 REFERENCE:

ICH Q7A, GMPs for Active Pharmaceutical Ingredients

ICH Q9, Quality Risk Management

ICH Q10, Pharmaceutical Quality System

WHO, Deviation handling and Quality Risk Management.

PICS, GMP Guide Part 1

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EU Good Manufacturing Practice- Chapter 1 Pharmaceutical Quality Systems

SOP-GMP-QA-0060 – Electronic Document management System

10.0 REVISION HISTORY:

Version no.	Effective Date	Description of Change
7.0	**24-Feb-2026**	**Reference change control Number: PR# 128239** 1. Scope modified as Internal quality audit & Continuous Improvement and Training Department. 2. Responsibility section modified to be in line with tabular world class format. 3. Under section 7.8 Included instruction 7.8.4 QA reviewer/approver shall ensure that the client's signature approval is obtained in the investigation report before closing the record in the Trackwise system, if required by the quality agreement.
6.0	**30-Jun-2025**	**Reference change control Number: PR#119594** 4. 7.5.4.1 Procedure for Process mapping is elaborated for better clarity. 5. Section no. 7.5.12 newly included about the procedure to handle the investigations when the root cause of nonconformity is attributable to individual error and the individual responsibility is no longer available in the organization. 6. Point 7.10.2 modified as: CQA shall select closed investigations from each operating unit monthly as per the below-mentioned selection criteria (but not limited to): • Critical deviations. • Re-classified deviations as per the escalation matrix criteria.

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		• Human error as root cause of deviation • Major customer complaints. • Late phase, commercial phase and major deviations. • Late phase & commercial phase product-related investigations of OOS & OOT • Recurrent OOS investigations of environmental monitoring and water analysis etc. • Deviations flagged during audits or regulatory inspections Note: Final classification shall be verified for correctness by CQA for all the closed investigations. 7. 7.10.5 and 7.10.6 sections are included about usage of formats for surveillance observations and responses through EDMS. 8. 7.12 – Procedure for document preparation in EDMS is added newly. 9. Throughout the procedure, sentences have been modified in line with English language proficiency requirements, and editorial changes have been made for better clarity.
5.0	**04-Jul-2024**	**Reference change control Number: PR#101764** The section 7.2.1 modified to add details of 5W1H in the description of a nonconformity. 1. Section No. 7.3.2 modified for better clarity on documentation of Sequence of events. 2. The point No. 7.3.3 "Problem shall be depicted in elaborative manner with process flow (pictorial diagram/graphs etc.), reference of documents and pictures etc., if applicable." removed as a part of simplification process. 3. Step No. 7.4.1.1 modified for better clarity.

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		4. In step No. 7.4.3.1 is updated to provide rationale only in case of review period selection for less than 12 months for historical check. 5. In the step No. 7.6.4 Note, "Provide detailed justification, if any of the above assessments do not apply or are not relevant" removed as a part of simplification process. 6. Section No. 7.10.3 modified to add acknowledgement of OUQA in the Template for observations of investigation surveillance. In the step No. 7.11.1 "Populate the template with reference to the table below." Removed repeated information along with the table content.

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